Quantification: 699_HYP_HP_z20c1-2
==================================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
----

measure	value
lacuna count	24
interior lacuna count	20
frame-edge lacuna count	4

Interior lacunae only: mean, median and sample SD (n = 20)
----

measure	kind	mean	median	sd	units
area_px2	lacuna shape	1596.45	1159	1500.33	px^2
major_axis_length_px	lacuna shape	75.767	62.3535	42.5431	px
minor_axis_length_px	lacuna shape	32.0835	30.2672	16.5017	px
aspect_ratio	lacuna shape	2.6077	2.2958	1.2153	unitless
eccentricity	lacuna shape	0.8563	0.8997	0.1175	unitless
solidity	lacuna shape	0.7509	0.76	0.0998	unitless
roots_count	headline	5.9	5	2.8266	count
ring_length_r30_px	headline	206.6	202.5	117.744	px
ring_length_r60_px	reported	392.9	351.5	243.024	px
owned_length_px	ownership-dependent	398.684	321.445	406.924	px
edge_count	ownership-dependent	22.55	18	22.3947	count
mean_edge_length_px	ownership-dependent	17.2249	16.906	5.3971	px
perimeter_px	normalised	233.335	177.631	136.358	px
ring_area_r30_px2	normalised	7126.7	6436	3382	px^2
ring_area_r60_px2	normalised	15727.5	14870.5	6982.25	px^2
in_frame_fraction_r30	normalised	0.9807	1	0.0611	unitless
in_frame_fraction_r60	normalised	0.9698	1	0.0936	unitless
ring_density_r30	normalised	0.0316771	0.0310795	0.0178668	px^-1
ring_density_r60	normalised	0.0261437	0.0272453	0.0148858	px^-1
roots_per_100px_perimeter	normalised	3.0611	3.3468	1.7137	per 100 px
ring_attached_length_r30_px	network v2	172.85	165.5	110.569	px
ring_attached_length_r60_px	network v2	283	224	202.172	px
ring_length_w_r30_px	network v2	229.857	221.661	133.443	px
ring_length_w_r60_px	network v2	438.293	390.063	273.767	px
sholl_crossings_r10	network v2	4.8	5	2.9128	count
sholl_crossings_r20	network v2	5.4	6	3.4702	count
sholl_crossings_r30	network v2	6	5.5	3.8798	count
loop_count	loops, report only	0.75	0	1.8317	count
loop_edge_share	loops, report only	0.0733	0	0.1346	share
ring_width_mean_r30_px	width	6.8335	6.7604	0.8725	px

Per field
----

measure	value	units
analysed_area_px2	991897	px^2
total_skeleton_length_px	17708	px
canalicular_length_density_per_px	0.0178527	px^-1
junction_count	354	count
junction_density_per_px2	0.00035689	px^-2
lacuna_count	24	count

lacunae_per_px2	2.42e-05	px^-2
skeleton_component_count	327	count
mean_component_length_px	54.1529	px
median_component_length_px	19	px
field_length_density_w_per_px	0.0199651	px^-1
field_density_without_flagged_per_px	0.0168374	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	1026	px
unattached_component_count	51	count
canalicular_length_density_attached_per_px	0.0178527	px^-1
field_length_density_with_unattached_per_px	0.018887	px^-1
loop_count	29	count
loop_edge_share	0.1184	share
edge_count	1022	count

Canalicular width, the new measure

measure	value	units
width_mean_px	6.9743	px
width_median_px	7.2111	px
width_p10_px	4	px
width_p90_px	10	px
width_pixels_used	13493	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	582	37.2403	23.3143	1.5973	0.7798	0.8362	-1.3908	9.6323	718.978	True
2	556	37.5638	24.5507	1.5301	0.7569	0.7596	0.3365	28.2194	661.201	False
3	701	53.6823	22.5905	2.3763	0.9071	0.7242	1.3601	138.937	341.827	False
4	1079	46.6189	34.6236	1.3464	0.6696	0.7599	-0.6851	205.172	387.171	False
5	3222	86.3749	58.8691	1.4672	0.7318	0.7263	-0.9453	298.969	636.32	False
6	4692	101.01	74.3902	1.3578	0.6765	0.6687	0.5065	385.336	36.169	True
7	12231	202.474	100.48	2.0151	0.8682	0.7797	0.0478	526.667	38.6221	True
8	662	43.8835	32.7282	1.3408	0.6662	0.6228	-1.315	369.021	352.307	False
9	1329	56.9074	38.8202	1.4659	0.7312	0.7408	-0.6935	386.281	385.088	False
10	574	63.7	12.3761	5.147	0.9809	0.8658	-1.4148	410.084	638.57	False
11	428	52.3766	12.3555	4.2391	0.9718	0.7698	-0.7977	531.266	620.591	False
12	961	72.5694	18.5814	3.9055	0.9667	0.8076	1.4502	539.664	699.95	False
13	1511	109.373	23.0638	4.7422	0.9775	0.6218	1.4725	559.445	447.167	False
14	434	34.6654	18.7525	1.8486	0.8411	0.7601	1.3394	579.993	424.74	False
15	599	49.8148	18.0062	2.7665	0.9324	0.7799	1.3874	565.563	692.351	False
16	898	39.9158	30.7419	1.2984	0.6378	0.8635	1.334	631.296	609.345	False
17	2133	94.4671	56.7385	1.665	0.7995	0.538	0.9278	716.599	755.109	False
18	2065	77.2677	36.7006	2.1054	0.88	0.8765	-0.6191	783.319	730.515	False
19	2136	110.111	32.0301	3.4377	0.9568	0.7495	-0.4861	842.146	688.921	False
20	7245	127.692	88.342	1.4454	0.7221	0.8447	-1.2489	821.694	971.904	True
21	7049	221.75	78.7671	2.8153	0.9348	0.5533	-1.5418	863.529	297.474	False
22	2325	112.289	29.7925	3.769	0.9642	0.7992	1.5515	897.959	158.373	False
23	2028	91.0023	34.0428	2.6732	0.9274	0.8145	0.9697	910.357	523.35	False
24	1239	61.0069	27.5385	2.2153	0.8923	0.8844	-1.1926	1001.21	883.603	False

Network per lacuna

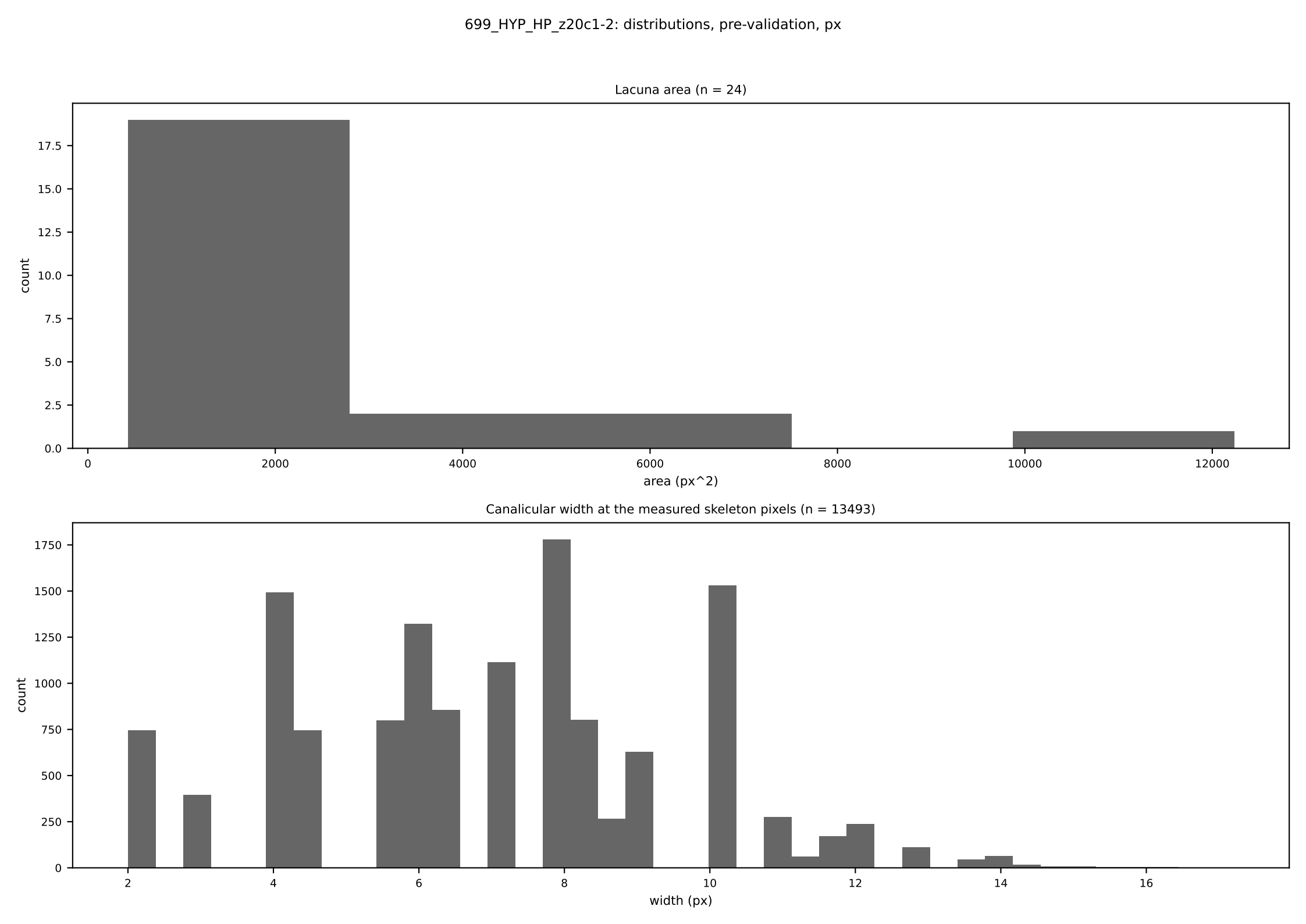
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	1	89	171	17.4853	1	17.4853	6.933
2	5	213	313	378.635	26	14.5629	6.717
3	12	320	421	401.321	30	13.3774	7.496
4	8	218	510	563.617	26	21.6776	8.6503
5	8	343	755	809.838	50	16.1968	6.2878
6	9	347	687	1878.48	104	18.0623	7.0427
7	11	254	275	296.108	18	16.4504	7.1579
8	5	277	638	1661.83	91	18.2618	7.9314
9	8	324	718	297.652	19	15.6659	6.946
10	6	192	424	345.238	17	20.3081	6.6589
11	6	256	390	539.617	35	15.4176	7.3876
12	4	180	267	80.6985	8	10.0873	6.2367
13	12	433	781	951.222	54	17.6152	7.7179
14	4	81	184	25.3137	3	8.4379	6.2785
15	5	132	165	166.255	8	20.7819	7.2694
16	7	348	665	572.902	25	22.9161	6.0891
17	8	112	254	103.77	7	14.8242	6.8038
18	2	40	40	52.6274	4	13.1569	5.7532
19	5	73	124	46.2132	5	9.2426	6.3638
20	1	26	67	19.4853	1	19.4853	5.3649
21	5	330	688	680.801	31	21.9613	7.6111
22	3	121	254	120.284	6	20.0474	7.602
23	3	104	205	157.368	5	31.4735	5.7226
24	2	35	62	18.4853	1	18.4853	5.146

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	115.083	2908	6188	0.6007	0.566	0.0306052	0.0276341	0.8689
2	125.539	4509	10030	0.8517	0.729	0.0472389	0.0312064	3.9828
3	169.681	6393	14892	1	1	0.0500548	0.0282702	7.0721
4	185.581	6924	15979	1	1	0.0314847	0.0319169	4.3108
5	418.517	11182	24664	1	1	0.0306743	0.0306114	1.9115
6	534.86	8448	16501	0.7334	0.6731	0.0410748	0.0416338	1.6827
7	728.541	11456	21516	0.6459	0.6063	0.0221718	0.0127812	1.5099
8	162.51	4532	11047	1	1	0.0611209	0.0577532	3.0767
9	221.179	5944	14647	1	1	0.0545088	0.0490203	3.617
10	140.326	6761	15870	1	1	0.0283982	0.0267171	4.2758
11	121.539	5538	10494	1	1	0.0462261	0.0371641	4.9367
12	165.74	4908	10686	1	1	0.0366748	0.024986	2.4134
13	316.149	8606	20213	1	1	0.0503137	0.0386385	3.7957
14	106.811	2715	6625	1	1	0.0298342	0.0277736	3.7449
15	131.439	3833	7002	1	1	0.0344378	0.0235647	3.8041
16	132.267	6479	14849	1	1	0.053712	0.0447842	5.2923
17	363.747	9538	20842	1	1	0.0117425	0.0121869	2.1993
18	224.528	4228	7038	1	1	0.00946074	0.00568343	0.8908
19	285.22	8488	19536	1	1	0.00860038	0.00634726	1.753
20	652.181	10802	23498	0.7216	0.6773	0.00240696	0.00285131	0.1533
21	676.281	17978	35439	1	1	0.0183558	0.0194136	0.7393
22	308.551	9405	21041	1	1	0.0128655	0.0120717	0.9723
23	247.421	8982	20148	1	1	0.0115787	0.0101747	1.2125
24	163.681	5591	13507	0.7628	0.6662	0.00626006	0.00459021	1.2219

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	56	56	99.8406	189.539	3	2	2
2	147	197	231.024	337.451	4	6	4
3	295	348	355.25	468.576	8	8	6
4	205	428	244.238	573.345	5	7	8
5	216	332	386.433	843.11	9	9	12
6	306	474	386.35	769.671	8	5	8
7	248	269	287.037	308.764	7	7	3
8	277	619	309.48	708.76	6	6	6
9	302	597	354.38	799.772	6	7	8
10	184	251	212.297	473.505	3	6	4
11	249	354	296.037	442.676	7	9	9
12	81	81	189.912	287.581	5	4	4
13	431	610	496.688	891.323	12	10	10
14	74	119	92.0833	215.551	3	1	3
15	132	155	141.255	175.912	5	4	2
16	298	551	387.25	745.428	6	12	13
17	112	182	120.498	274.238	4	2	5
18	40	40	42.5563	42.5563	1	1	0
19	73	91	82.7696	147.024	1	2	2
20	21	21	29.1421	74.7696	1	0	2
21	190	450	369.049	768.884	6	9	13
22	79	160	129.083	273.853	3	3	6
23	48	71	116.397	225.48	2	2	4
24	24	24	40.4558	70.8406	0	0	1


699_HYP_HP_z20c1-2: distributions, pre-validation, px
Lacuna area (n = 24)
count
area (px^2)
0
2000
4000
6000
8000
10000
12000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Canalicular width at the measured skeleton pixels (n = 13493)
count
width (px)
0
250
500
750
1000
1250
1500
1750
2
4
6
8
10
12
14
16